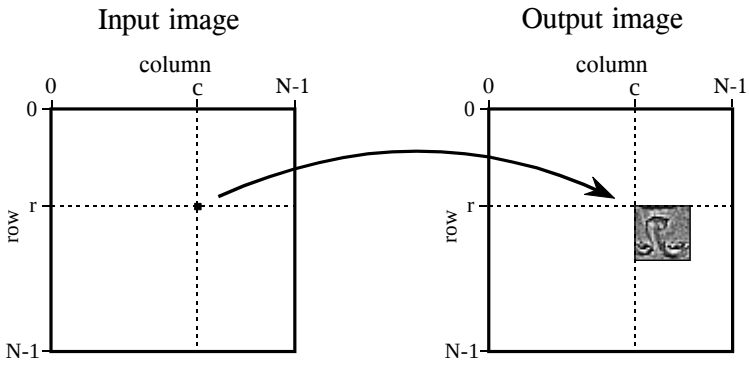

FIGURE 24-14
Image convolution viewed from the input side. Each pixel in the input image contributes a scaled and shifted PSF to the output image. The output image is the sum of these contributions. The face is inverted in this illustration because this is the PSF we are using.

be confused by the face appearing upside down in this figure; this upside down face *is* the PSF we are using in this example (Fig. 24-13a). In the input side view, there is no rotation of the PSF, it is simply shifted.

Image convolution viewed from the output is illustrated in Fig. 24-15. Each pixel in the output image, such as shown by the sample at [r,c], receives a contribution from many pixels in the input image. The PSF is rotated by 180° around pixel [0,0], and then shifted such that pixel [0,0] in the PSF is aligned with pixel [r,c] in the input image. If the PSF only uses positive indexes, it will be to the upper-left of pixel [r,c] in the input image. The value of the pixel at [r,c] in the output image is found by multiplying the pixels in the rotated PSF with the corresponding pixels in the input image, and summing the products. This procedure is given by Eq. 24-3, and in the program of Table 24-1.

EQUATION 24-3
Image convolution. The images $x[\ ,\ ]$ and $h[\ ,\ ]$ are convolved to produce image, $y[\ ,\ ]$. The size of $h[\ ,\ ]$ is $M \times M$ pixels, with the indexes running from 0 to $M-1$. In this equation, an individual pixel in the output image, $y[r,c]$, is calculated according to the output side view. The indexes j and k are used to loop through the rows and columns of $h[\ ,\ ]$ to calculate the sum-of-products.

$$y[r,c] = \sum_{k=0}^{M-1} \sum_{j=0}^{M-1} h[k,j]\, x[r-k, c-j]$$

Notice that the PSF rotation resulting from the convolution has undone the rotation made in the design of the PSF. This makes the face appear upright in Fig. 24-15, allowing it to be in the same orientation as the pattern being detected in the input image. That is, we have successfully used *convolution* to implement *correlation*. Compare Fig. 24-13c with Fig. 24-15 to see how the bright spot in the correlation image signifies that the target has been detected.